

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash, HTML/CSS, Flask

Tools/Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Pandas, FastAPI, Docker, Kubernetes, AWS SageMaker, MLflow, Git, CI/CD, Airflow, LLM Frameworks

Competence: Deep Learning, Machine Learning, Computer Vision, MLOps, Data Science, Time Series Prediction, Segmentation, Perception, Model Deployment, Edge AI, Large Language Models (LLMs), Reinforcement Learning, Signal Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored and implemented novel architectures and physics-informed hybrid loss functions to advance model accuracy and stability for complex scientific datasets.
- Collaborated with PhD researchers on deep learning for metrology, contributing to a publication presented at the FLOMEKO conference.
- Developed interactive dashboards for real-time data visualization and comprehensive model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced vitiligo segmentation accuracy to 94% and optimized inference speed by 5x using advanced AI algorithms, deployed with Docker and FastAPI.
- Developed a real-time tennis foul detection system leveraging TensorRT, reducing compute cost and processing time by 6x for edge deployment.
- Managed robust MLOps pipelines on AWS SageMaker, proactively addressing YOLO model data drift and maintaining Kubernetes infrastructure for continuous delivery.
- Developed Proof-of-Concepts (POCs) for clients using state-of-the-art YOLO, few-shot learning, and CNN architectures for diverse perception tasks.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed an AI chatbot application utilizing OpenAI API and advanced LLM frameworks, integrating seamlessly into existing web services.
- Contributed to full-stack development of an AI-powered service website using Flask, HTML, and CSS, ensuring responsive design and user experience.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented robust ETL pipelines and web scrapers for efficient ingestion and processing of structured and unstructured data.
- Automated critical data ingestion, cleaning, and transformation workflows, delivering clean datasets for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a high-performing weighted-probability ensemble model, integrating DPFunc and DeepGOPlus, for protein function prediction.
- Utilized AlphaFold for advanced feature extraction and implemented taxonomic bias mitigation strategies, achieving a 4% increase in model accuracy across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a thesis project on real-time Bitcoin price forecasting, applying modern ML techniques for time series prediction and analysis.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to develop Saaz, the world's first AI-driven self-playing Chitralli Sitar, integrating signal processing with music generation LLMs.
- Project gained significant recognition, featured on Coke Studio in a music video by Hassan Raheem.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, contributing to open-source ML knowledge.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized coursework in Deep Learning, Computer Vision, Reinforcement Learning, and Machine Learning Engineering.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Thesis: Razanlytics: Real-time Bitcoin Price Forecasting.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python